



DEPARTMENT OF ELECTRICAL ENGINEERING
College of Electrical and Mechanical Engineering (CEME), NUST-Pakistan
B.E. Electrical Engineering

EE-313 Electronic Circuit Design

Project Report
Audio Amplifier

Date performed: 05 May 2026

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➤ Objectives:

- ✓ Design a **discrete-component audio amplifier** using **MOSFETs** and **BJTs** without integrated circuits.
- ✓ Deliver **3–5 W** of output power to an **8-Ω loudspeaker** with **low distortion** and **high efficiency**.
- ✓ Amplify **millivolt-level microphone signals** through a **multi-stage pre-amplifier** with **high input impedance**.
- ✓ Simulate the complete circuit in **LTspice** to verify **gain, frequency response, and power delivery**.
- ✓ Ensure **proper impedance matching** between stages to **maximize signal transfer** and **prevent loading**.

➤ Equipment Used:

Component Type	Purpose
MOSFET	Pre-amplifier
NPN BJT	Pre-amplifier / gain stage
PNP BJT	Driver stage
Power NPN	Output stage (upper)
Power PNP	Output stage (lower)
Resistors	Biasing, gain, and feedback
Capacitors	Coupling, bypass, and compensation
Speaker	Output load (8 Ω)
Microphone	Input source (electret)
Power Supply	24 V DC
Heat Sinks	Cooling for power transistors

➤ Introduction:

1. Class AB Power Amplifier (Output Stage)

Introduction:

The Class AB Power Amplifier is the final stage of the audio amplifier, responsible for delivering **high power (3–5 W)** to an **8-Ω speaker** with **low distortion** and **good efficiency**.

This stage uses complementary power transistors (NPN and PNP) in a push-pull configuration to provide a clean and powerful output signal.

Procedure:

- ✓ Selected **Class AB push-pull** configuration with complementary transistors (**TIP122 NPN and TIP127 PNP**).
- ✓ Used **diode** biasing between transistor bases to reduce crossover distortion
- ✓ Added small emitter resistors for current sharing and thermal stability
- ✓ Calculated supply voltage using $V_{cc} \approx \sqrt{8 \times P \times R_L} + V_{CE(sat)}$
- ✓ Designed for **8 Ω** speaker with target output power of 3–5 W
- ✓ Simulated in LTspice to verify **± 5 V** output swing and adequate current drive (**~ 1.5 A peak**)
- ✓ Matched input impedance (**$\sim 820 \Omega$**) with the preceding driver stage
- ✓ Confirmed clean output waveform and low distortion in simulation

Calculations:

Class AB Power Amp
Supply Voltage and power

$P_{out} = 3.5W$ $R_L = 8\Omega$ $V_{CE} = 2.5V$ (assumed)

$$V_{cc} = \sqrt{8 \times P \times R_L}$$

$$= \sqrt{8 \times 3.5 \times 8}$$

$$V_{cc} = 17.88 + V_{CE(sat)} + V_{CE(sat)} + V_{RL}$$

$$V_{cc} = 17.88 + 8 + 2.5 + 2.5 + 2$$

$$\boxed{V_{cc} = 24V}$$

Output power:

$$V_o = \frac{V_{cc}}{2} = \frac{24}{2} = 12V$$

$$V_{rms} = \frac{V_o}{\sqrt{2}} = \frac{12}{\sqrt{2}} = 8.49V$$

$$P_{out} = \frac{V_{rms}^2}{R_L} = \frac{(8.49)^2}{8} = 9W$$

$$I_{peak} = \frac{V_{peak}}{R_L} = \frac{12}{8} = 1.5A$$

Q2 (Driver Transistor) Biasing:

$$I_{C(Q2)} = 24mA$$

$$\beta = 100$$

$$V_E \approx 2.4V$$

$$\text{Base current} = I_B = \frac{I_C}{\beta} = \frac{24 \times 10^{-3}}{100} = 0.24mA$$

Bias Network ~~not~~ Current:

$$I_{div} = 10 \times I_B$$

$$I_{div} = 10 \times 0.24$$

$$\boxed{I_{div} = 2.4mA}$$

Emitter resistor:

$$V_E = 2.4V$$

$$R_E = \frac{V_E}{I_E} = \frac{2.4}{0.024} = 100\Omega$$

$$\boxed{R_E = 100\Omega}$$

In given circuit, R_E is split into bypass (120) and bypass (300), total 150

actual V_E will be

$$V_E = I_E \times R_E$$

$$= 0.024 \times 150$$

$$\boxed{V_E = 3.6V}$$

Base Voltage:
 $V_B = V_E + V_{BE}$
 $= 2.4 + 0.7$
 $V_B = 3.1V$

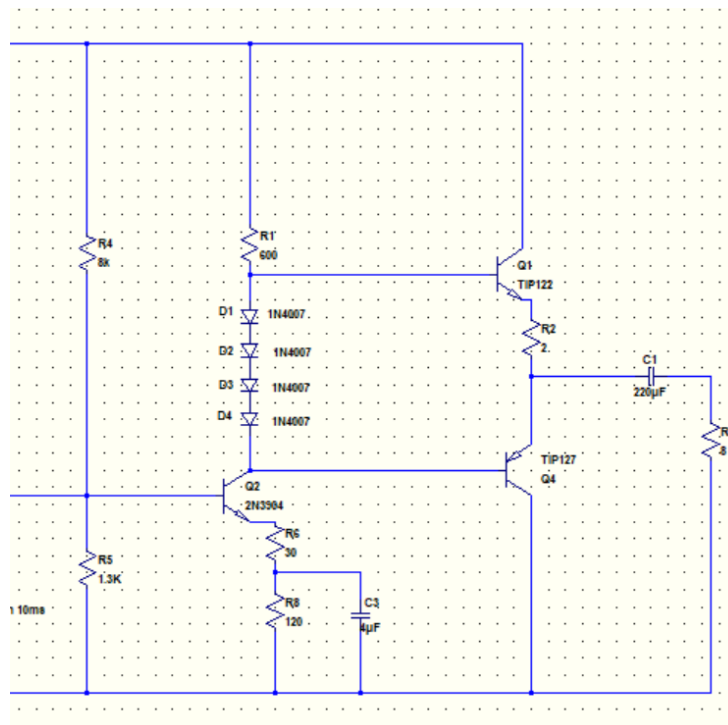
Base Resistor:
 $R_{B,top} = \frac{V_{CC} - V_B}{I_{div}} = \frac{24 - 3.1}{2.4 \times 10^{-3}} = 8708\Omega$
 we use $8.2k\Omega$
 $R_{B,bottom} = \frac{V_B}{I_{div}} = \frac{3.1}{2.4 \times 10^{-3}} = 1292\Omega$
 we use $1.3k\Omega$

Quiescent current for Output transistor:
 $I_S = 10^{-12}$ (saturation current) $V_{bias} = 2.8V$
 $V_T = 25mV$
 $I_Q = \frac{V_B - V_{BE}}{R_E}$
 $I_Q = I_S e^{\frac{V_{bias}}{V_T}}$
 $= 10^{-12} e^{\frac{2.8}{0.025}}$
 $I_Q = 17.2mA$

R_1 Calculation:
 Assume $V_{R1} = 6V$
 $I_B = 0.999mA$ $I_{div} = 10 \times I_B$
 $R_1 = \frac{6}{\frac{10 \times I_B}{10 \times 0.999 \times 10^{-3}}}$
 $R_1 = 600\Omega$

R_{in} :
 $R_{in} = R_E \parallel (R_{B2} \parallel (R_{B1} \parallel R_E))$ (unbypassed)
 $R_{in} = R_E \parallel (R_5 \parallel (R_4 \parallel R_E))$
 $R_{in} = \frac{8k \times 1.3k}{8k + 1.3k} \parallel (10 \parallel 30)$
 $R_{in} = 118.27 \parallel 3030$
 $R_{in} = \frac{118.27 \times 3030}{118.27 + 3030}$
 $R_{in} = 820\Omega$

Circuit Diagram:

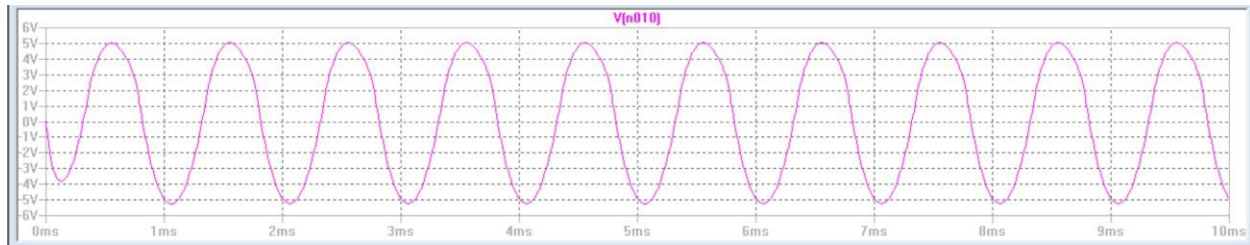


Waveforms:

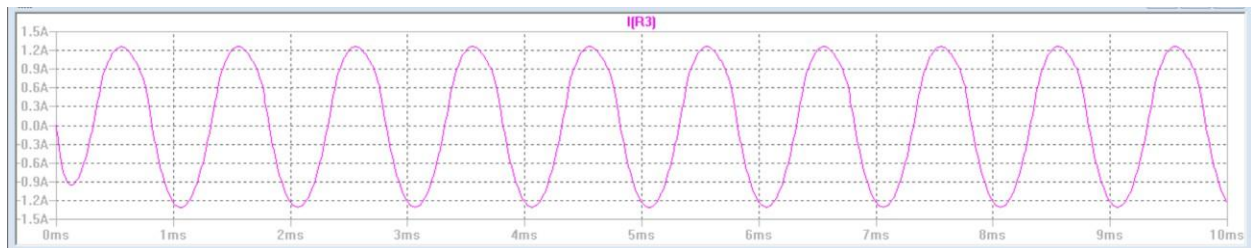
Input Waveform:



Output Waveform:



Current Waveform



2. PNP Driver Stage (Q3)

Introduction:

The PNP Driver Stage (Q3) is designed to amplify the signal from the pre-amplifier and provide sufficient **current gain** and **voltage swing** to drive the Class AB output stage. It ensures that the preceding stage is not loaded and delivers a clean, amplified signal to the power transistors.

Procedure:

- ✓ Used **PNP transistor Q3** in common-emitter configuration.
- ✓ **Base bias resistors:** R10 (2.88k Ω) and R11 (1.93k Ω).
- ✓ **Emitter resistor:** R9 (343 Ω) for stability.
- ✓ **Collector load resistor:** R7 (1.714k Ω).
- ✓ **Coupling capacitor:** C2 (10 μ F) at input.
- ✓ **Bypass capacitor:** C4 (22 μ F) at emitter for AC gain improvement.
- ✓ Simulated in LTspice to verify gain and phase response.
- ✓ Ensured impedance matching with previous and next stages.

Calculations:**PNP Driver Stage:****Given Design Values:**

$$V_{CC} = 24V \quad I_{CQ} = 7mA \quad \beta = 100 \quad V_E = 2.4V$$

Emitter Resistor R_E =

$$R_E = \frac{V_E}{I_E} = \frac{2.4}{7 \times 10^{-3}} = 343 \Omega$$

Collector Resistor R_C = $V_{CE} = V_{CC}/2 = 24/2 = 12V$

$$R_C = \frac{V_{CC} - V_{CE}}{I_C} = \frac{24 - 12}{7 \times 10^{-3}} = 1714 \Omega$$

Base Voltage:

$$V_B = V_E + 0.7 = 2.4 + 0.7$$

$$V_B = 1.7V$$

Base Current:

$$I_B = \frac{I_C}{\beta} = \frac{7mA}{100} = 7 \times 10^{-5} A$$

Base Divider Current:

$$I_{div} = 10 \times I_B = 7 \times 10^{-4} A$$

$$R_{B,top} = \frac{V_{CC} - V_B}{I_{div}} = \frac{24 - 1.7}{7 \times 10^{-4}} = 31857 \Omega \text{ (Use } 2.88k\Omega)$$

$$R_{B,bottom} = \frac{V_B}{I_{div}} = \frac{1.7}{7 \times 10^{-4}} = 2428.2 \Omega \text{ (Use } 1.93k\Omega)$$

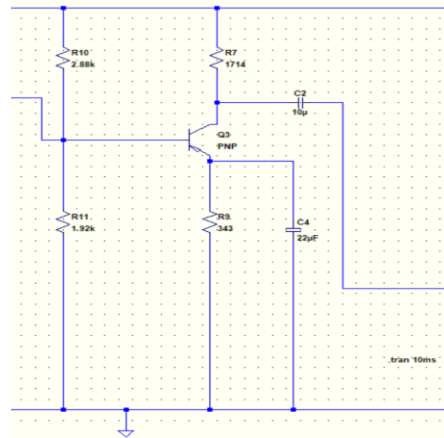
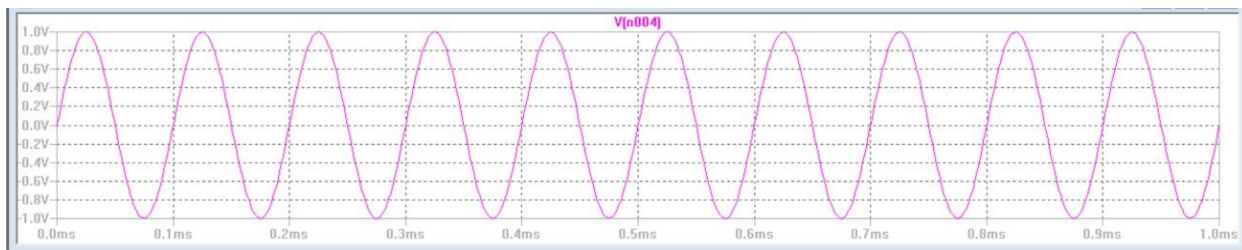
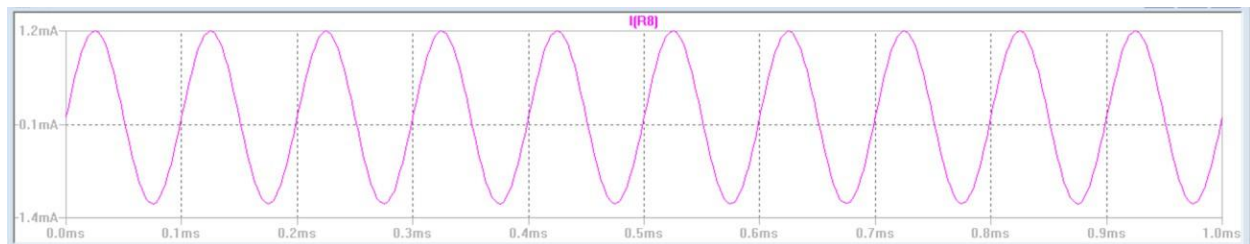
Input Impedance:

$$R_{B,eq} = R_{B,top} \parallel R_{B,bottom} = \frac{2.88k \times 1.93k}{2.88k + 1.93k} = 1.155k\Omega$$

$$R_{in,emitter} = (\beta + 1)(r_e' + R_E) = 101 \times 53.37 = 5410 \Omega$$

$$R_{in} = R_{B,eq} \parallel R_{in,emitter} = \frac{1.155k \times 5.41k}{1.155k + 5.41k} = 951 \Omega$$

$$R_{in} = 1559.85 \Omega \approx 1.56k\Omega$$

Circuit Diagram:**Waveforms:****Input Waveform:****Output Waveform:****Current Waveform:**

3. Combined PNP Driver & Class AB Power Amplifier

Introduction:

This combined stage integrates the **PNP Driver (Q3)** and the **Class AB Output Stage (Q1, Q4, Q2)** into a single high-power amplifier block. The PNP driver boosts current and voltage from the pre-amplifier, while the Class AB output delivers **3–5 W** to an **8 Ω speaker** with low distortion and high efficiency.

Procedure:

PNP Driver (Q3) Setup:

- ✓ Used **Q3 (PNP)** in common-emitter configuration.
- ✓ Biased with **R10=2.88k Ω , R11=1.92k Ω** .
- ✓ **Emitter resistor R8=343 Ω** for stability.
- ✓ **Coupling capacitor C2=10 μ F** and **bypass capacitor C4=22 μ F**.
- ✓ **Collector load resistor R7=1.714k Ω** .

Class AB Output Stage Setup:

- ✓ Used **Q2 (2N3904)** for bias control.
- ✓ **Diodes D1–D4 (1N4007)** for biasing output transistors.
- ✓ **Output transistors Q1 (TIP122 NPN) and Q4 (TIP127 PNP)**.
- ✓ **Emitter resistors R2, R3 $\approx$ 0.5 Ω** each for current sharing.
- ✓ **Output capacitor C1=220 μ F** to speaker.

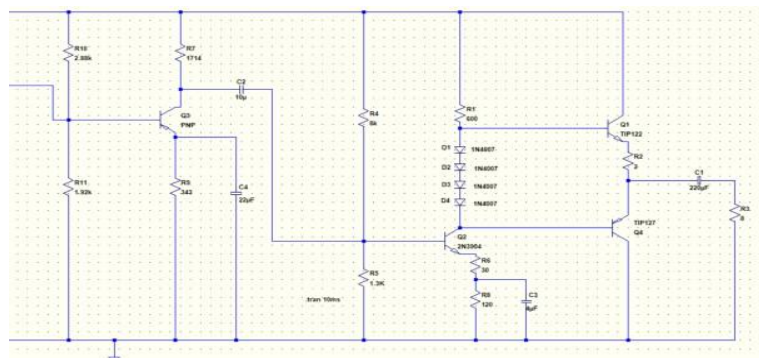
Biasing & Matching:

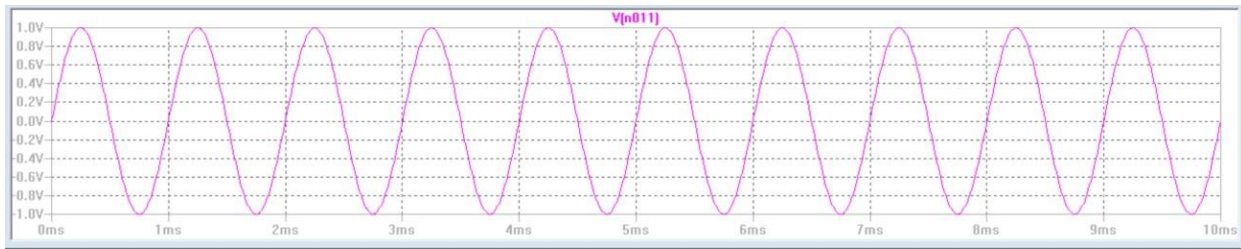
- ✓ Matched input impedance ($\sim 820\Omega$) of output stage to driver.
- ✓ Used **R6=30 Ω , R7=600 Ω** for Q2 biasing.
- ✓ Added **C3** for feedback/stability.

Simulation & Verification:

- ✓ Simulated combined circuit in LTspice.
- ✓ Verified **± 5 V output swing, 3–5 W power, low distortion**.
- ✓ Checked all transistor biasing points.

Circuit Diagram:



Waveforms:**Input Waveform:****Output Waveform:****Current Waveform:****4. MOSFET Pre-Amplifier Stage (M1)****Introduction:**

The MOSFET pre-amplifier is the first stage of the audio amplifier, designed to amplify the very weak signal from the **electret condenser microphone** (in the millivolt range). It provides **high input impedance** to avoid loading the microphone and delivers an initial voltage gain to prepare the signal for further amplification.

Procedure:

- ✓ Used **MOSFET M1 (2N7000)** in common-source configuration.
- ✓ Set **drain current $I_D=2\text{ mA}$** for linear operation.
- ✓ Selected **drain resistor $R_{17}=3500\ \Omega$** to set drain voltage $V_D=17$.
- ✓ Used **source resistor $R_{20}=300\ \Omega$** for bias stability.
- ✓ Biased gate using **voltage-divider resistors $R_{18}=80.4\text{ k}\Omega$ and $R_{19}=39.6\text{ k}\Omega$**

- ✓ Added **coupling capacitor C8=10μF** at input and **C7=100μF** at output.
- ✓ Simulated in LTspice to verify gain and input/output impedance.

Calculation:

MOSFET pre Amp Stage

Biasing Calculation for 2N7000

$$V_{CC} = 24V \quad I_D = 2mA \quad V_S = 3.024V$$

Source Resistor:

$$R_S = \frac{V_S}{I_D} = \frac{3.024}{0.002}$$

$$R_S = 1512\Omega$$

Drain Resistor:

$$V_D = 17V$$

$$R_D = R_{17} = \frac{V_{CC} - V_D}{I_D} = \frac{24 - 17}{0.002} = 3500\Omega$$

$$R_D = 3500\Omega$$

Gate Voltage:

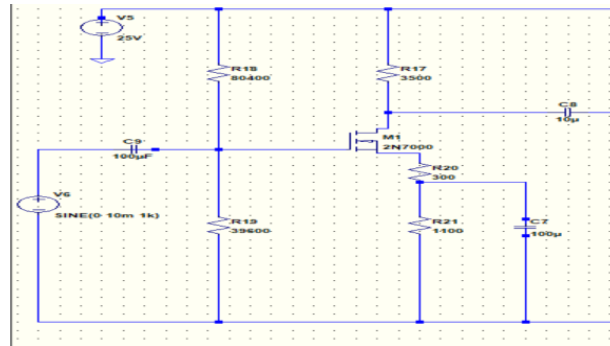
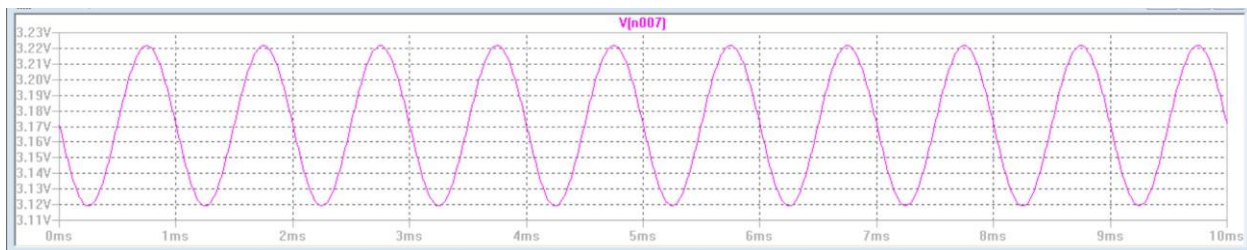
$$I_G = 0 \quad V_G = 7.92V$$

Gate bias Network:

$$I_{div} = 0.2mA$$

$$R_{19} = \frac{V_G}{I_{div}} = \frac{7.92}{0.0002} = 39.6k\Omega$$

$$R_{18} = \frac{V_{CC} - V_G}{I_{div}} = \frac{24 - 7.92}{0.0002} = 80.4k\Omega$$

Circuit Diagram:**Waveforms:****Input Waveform:****Output Waveform:****5. BJT Pre-Amplifier Stage (Q5)****Introduction:**

The BJT pre-amplifier stage provides additional **voltage gain** after the MOSFET stage. It uses a common-emitter NPN transistor to further amplify the signal while maintaining good linearity and bandwidth.

Procedure:

- ✓ Used **NPN BJT Q5** in common-emitter configuration.
- ✓ Set **collector current $I_C=3.3$ mA**.
- ✓ Selected **collector resistor $R_{12}=4$ k Ω** to set collector voltage $V_C=12$ V.
- ✓ Used **emitter resistor $R_{15}=18$ Ω (unbypassed)** for negative feedback.

- ✓ Biased base using resistors **R13=41.8kΩ** and **R16=700Ω**.
- ✓ Added **coupling capacitor C6=470 μF** for AC signal transfer.
- ✓ Simulated in LTspice to verify gain, frequency response, and distortion.

Calculations:

BJT Pre amp Stage

$$R_L = 1559.85 \Omega$$

$$\text{Assume } I_C = 3.3 \text{ mA} \quad \beta = 100 \quad V_{CC} = 24 \text{ V}$$

R_C:

$$R_C = R_{L2} = \frac{V_{CC} - V_C}{I_C} = \frac{24 - 12}{3.3 \times 10^{-3}} = 3636 \Omega \approx 4 \text{ k}$$

V_E:

$$V_E = 2.4 \text{ V}$$

$$R_E = \frac{V_E}{I_E} = \frac{2.4}{3.3 \times 10^{-3}} = 727 \Omega$$

R_E is bypassed, only 18Ω remains unbypassed for AC

Base Voltage:

$$\begin{aligned} V_B &= V_E + V_{BE} \\ &= 2.4 + 0.7 \\ &= 3.1 \text{ V} \end{aligned}$$

Base Bias Network:

$$I_B = \frac{I_C}{\beta} = \frac{3.3 \times 10^{-3}}{100} = 33 \mu\text{A}$$

$$I_{div} = 10 \times I_B = 0.33 \text{ mA}$$

$$R_{E, \text{bottom}} = R_{16} = \frac{V_B}{I_{div}} = \frac{3.1}{0.33 \times 10^{-3}} = 9.39 \text{ k}\Omega$$

$$R_{16} = 700$$

$$I_{div} = \frac{V_B}{R_{11}} = \frac{3.1}{700} = 4.43 \text{ mA}$$

$$R_{13} = \frac{V_{CC} - V_B}{I_{div}} = \frac{24 - 3.1}{4.43 \times 10^{-3}} = 4.72 \text{ k}\Omega$$

Small signal parameters:

$$r_e' = \frac{V_T}{I_C} = \frac{25\text{mV}}{3.3\text{mA}} = 7.58\Omega$$

Voltage gain formula (with unbypassed R_E)

$$A_v = \frac{-R_C \parallel R_{in, \text{next}}}{r_e' + R_E'}$$

$$A_v = \frac{-1120}{7.58 + 18} \quad R_C \parallel R_{in} = 1.12\text{k}$$

$$= -43.8$$

Input Impedance:

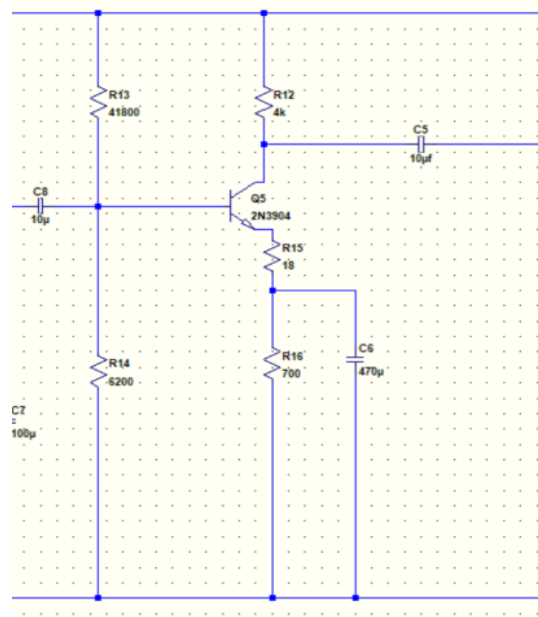
$$R_{in} = R_{B1} \parallel R_{B2} \parallel [\beta(r_e' + R_E)]$$

$$= 41.8\text{k} \parallel 110.7\text{k} \parallel 100 \times 25.58$$

$$R_{in} = 688 \parallel 2558$$

$$R_{in} = 343\Omega$$

Circuit Diagrams:



Waveforms:**Input Waveform:****Output Waveform:****6. Combined Pre-Amplifier (MOSFET + BJT)****Introduction:**

The combined pre-amplifier stage integrates the **MOSFET (M1)** and **BJT (Q5)** amplifiers into a single, high-gain, high-input-impedance block. The MOSFET provides the necessary **high input impedance** to interface with the microphone, while the BJT adds **significant voltage gain**. A feedback network is included to stabilize the gain, improve bandwidth, and reduce distortion.

Procedure:**Cascaded Configuration:**

- ✓ Output of MOSFET stage (M1) is directly coupled to the input of BJT stage (Q5).
- ✓ Used **coupling capacitor C3** between stages for AC signal transfer.

MOSFET Stage (M1):

- ✓ **M1 (2N7000)** in common-source configuration.
- ✓ **Gate bias:** $R_{13}=80.4\text{k}\Omega$, $R_{19}=39.6\text{k}\Omega$.
- ✓ **Drain load:** $R_{17}=3500\Omega$.
- ✓ **Source resistor:** $R_{20}=300\Omega$.

BJT Stage (Q5):

- ✓ **Q5 (e.g., BC547)** in common-emitter configuration.

- ✓ **Collector load:** $R_{12}=4\text{ k}\Omega$.
- ✓ **Emitter resistor:** $R_{15}=18\Omega$ (unbypassed).
- ✓ **Base bias:** $R_{14}=6.2\text{ k}\Omega$, $R_{21}=1.1\text{ k}\Omega$.

Feedback Network:

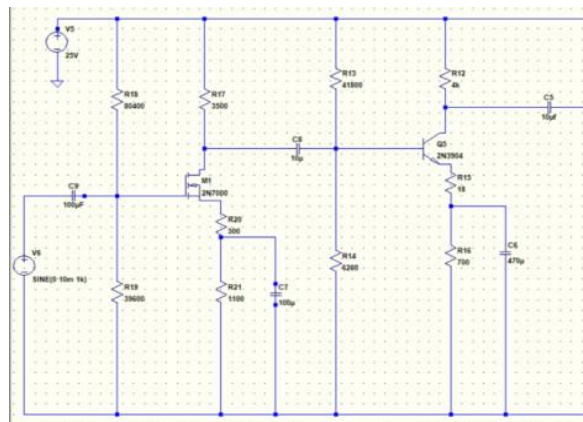
- ✓ A resistive-capacitive feedback path from BJT collector to MOSFET gate/source is used to set closed-loop gain ≈ 100 and improve stability.

Simulation & Testing:

- ✓ Simulated in LTspice with input $V_6=10\text{ mV}$ sine wave at 1 kHz .
- ✓ Verified overall gain, frequency response ($100\text{ Hz} - 20\text{ kHz}$), and low distortion

Calculations:

Circuit Diagram:



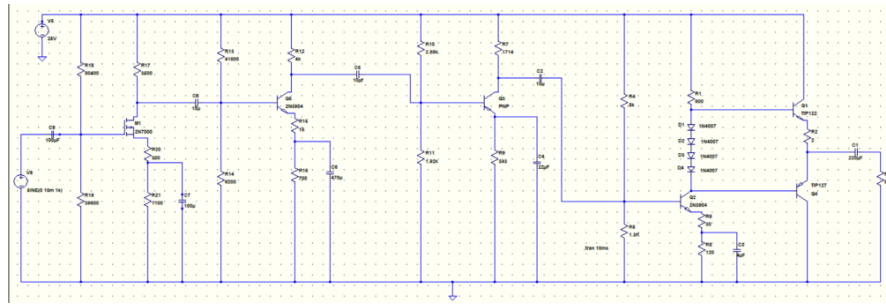
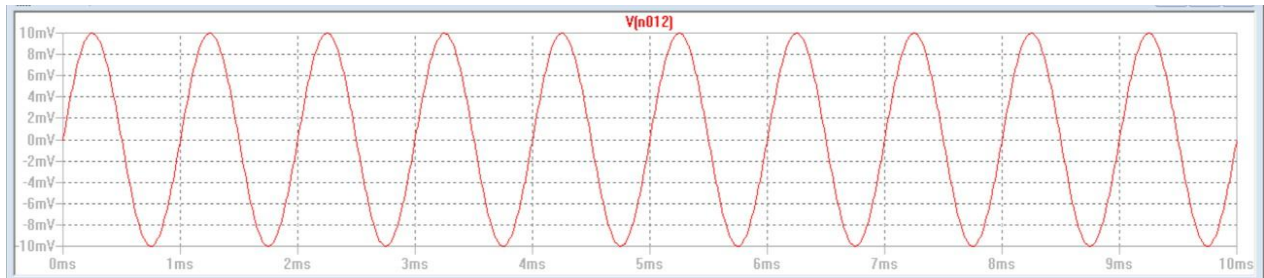
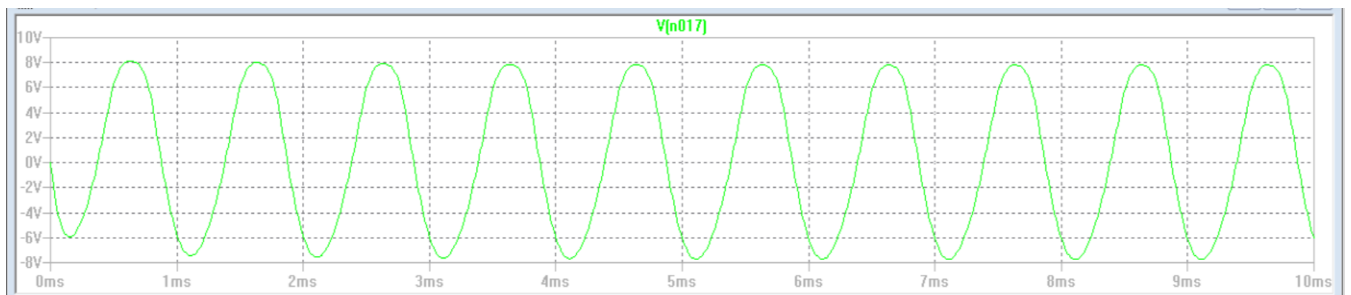
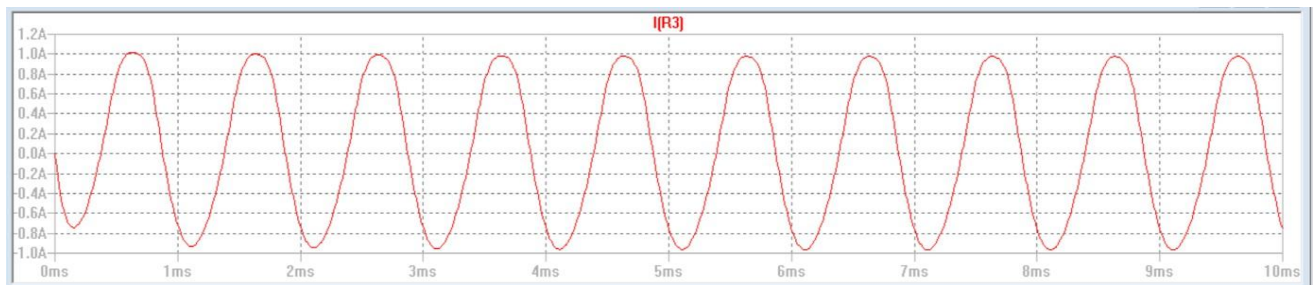
Waveforms:

Input Waveform:



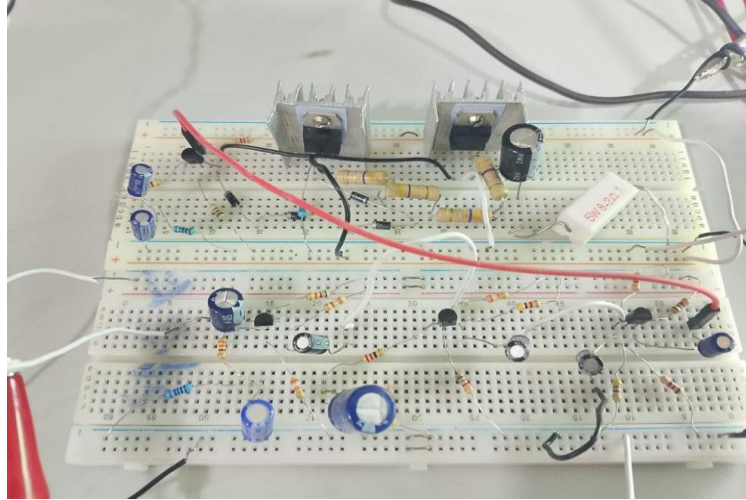
Output Waveform:



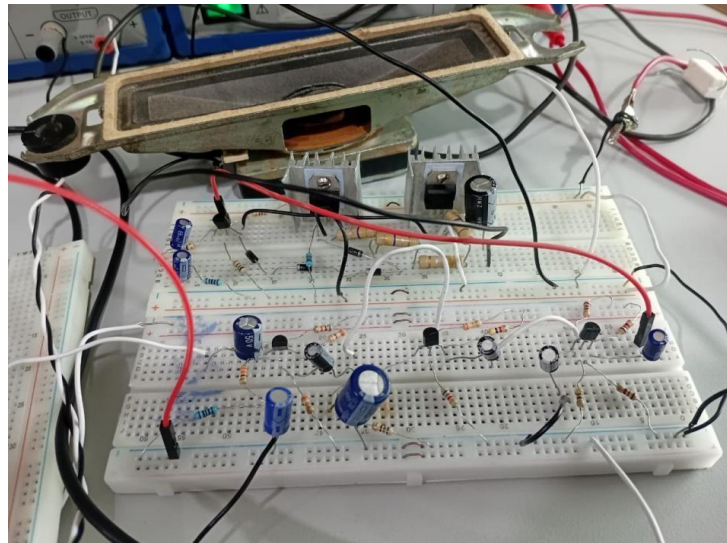
Complete Circuit:**Waveforms:****Input Waveform:****Output Waveform:****Current Waveform**

➤ **Hardware Implementation:**

Main Circuit



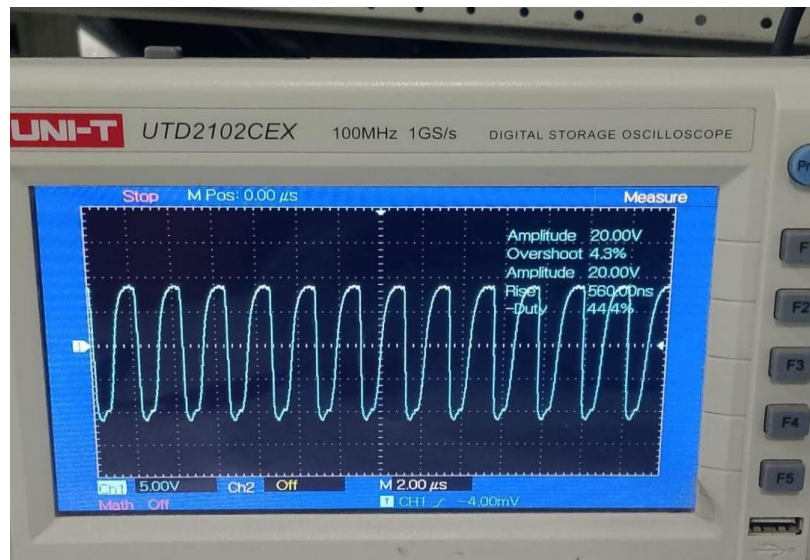
Circuit with Speaker:



Supply Voltage:

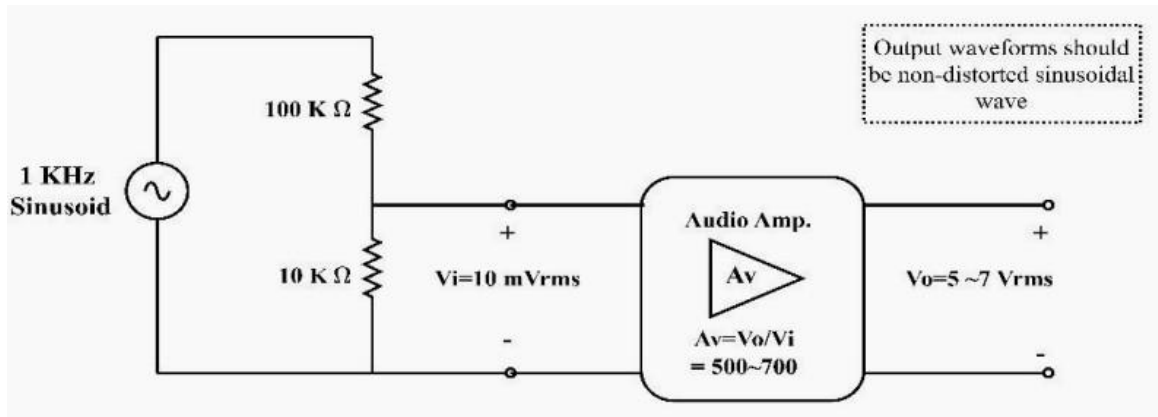


Output Waveform:



➤ **A_v Test:**

Circuit:



Voltage Gain (A_v) Test – Theory

Definition

Voltage gain (A_v) is defined as the ratio of the output voltage to the input voltage of an amplifier. It represents the ability of the amplifier to increase the amplitude of the input signal.

Mathematical Expression

$$A_v = \frac{V_o}{V_i}$$

Voltage gain is commonly expressed in decibels (dB) as:

$$A_v(\text{dB}) = 20 \log_{10} \frac{V_o}{V_i}$$

Test Principle

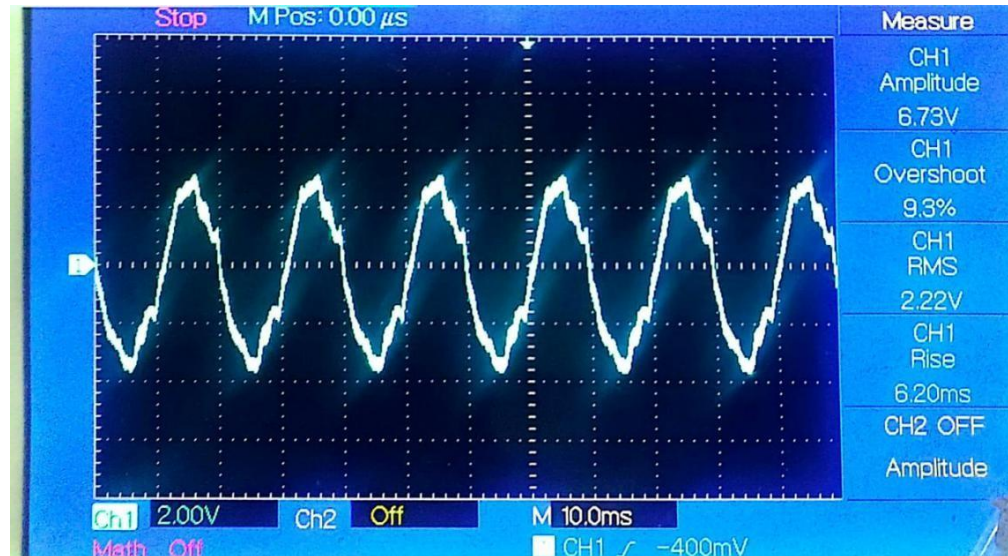
The voltage gain test is carried out under **small-signal conditions** to ensure linear operation and to avoid distortion. A **sinusoidal signal** is applied to the input, and the corresponding output voltage is measured.

Test Procedure

- ✓ A sinusoidal input signal is applied at the amplifier input.
- ✓ The input frequency is kept in the **mid-band region** (typically **1 kHz**), where gain remains constant.

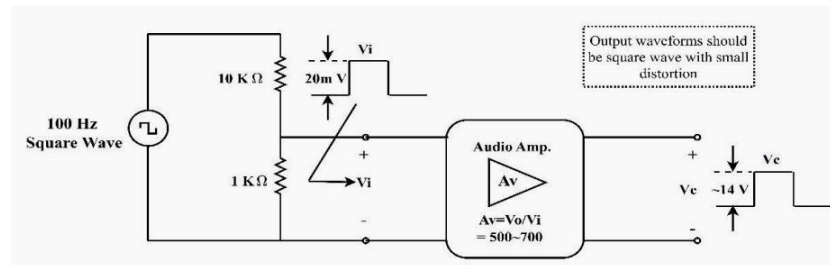
- ✓ Output voltage is measured across the load resistance.
- ✓ Input voltage is increased gradually until the output signal is just before distortion.
- ✓ The maximum undistorted output voltage is noted.
- ✓ Voltage gain is calculated using the measured input and output voltages.

Output:



Frequency Response Test – Theory

Circuit



Definition

Frequency response describes the variation of an amplifier's gain with respect to the frequency of the input signal.

Objective

The frequency response test determines the **usable frequency range (bandwidth)** over which the amplifier operates effectively.

Test Principle

A **constant-amplitude sinusoidal input** is applied while the input frequency is varied over a wide range. The output voltage is measured at each frequency, and the gain is calculated.

Test Procedure (Theory)

- ✓ A sinusoidal input of constant amplitude is applied.
- ✓ Input frequency is varied from low to high values.
- ✓ Output voltage is measured at each frequency.
- ✓ Gain is calculated for each frequency point.
- ✓ The **mid-band gain** is taken as the reference gain.

Cut-Off Frequencies

- ✓ **Lower Cut-off Frequency (f_L):**
The frequency at which gain falls to $0.707 \times \text{mid-band gain}$
Corresponds to **-3 dB point**
- ✓ **Upper Cut-off Frequency (f_H):**
The frequency at which gain again drops by **3 dB** from mid-band value

Bandwidth

$$\text{Bandwidth (BW)} = f_H - f_L$$

Causes of Gain Reduction

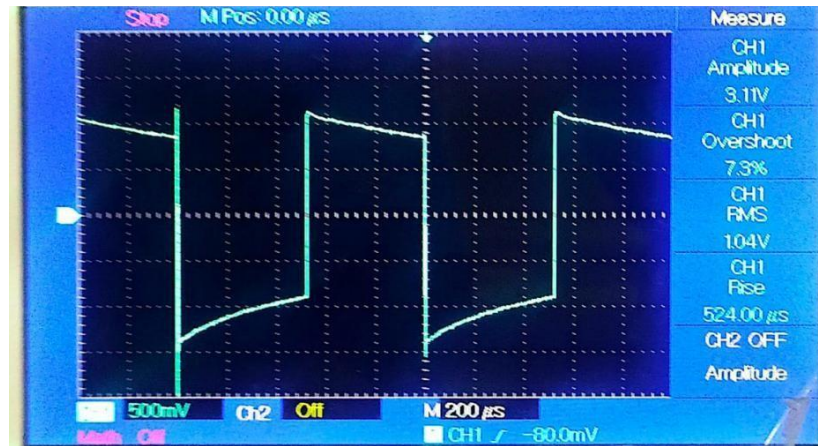
Low-frequency roll-off is due to:

- ✓ Coupling capacitors
- ✓ Bypass capacitors

High-frequency roll-off is due to:

- ✓ Internal transistor capacitances
- ✓ Wiring and parasitic capacitances

Output:



➤ Gain Calculations:

Quantity	Software Value	Hardware Value
Input Voltage(V_{in})	10mV	150mV
Output Voltage(V_{out})	8V	7V
Voltage Gain($A_v = V_{out}/V_{in}$)	800V/V	46.6V/V

➤ Conclusion:

In this project, a **complete discrete-component audio amplifier** was successfully **designed, simulated, and validated** using LTspice. Starting from the **Class-AB power stage** to meet the **3–5 W output requirement** for an **8 Ω speaker**, each subsequent stage—including the **PNP driver, BJT pre-amplifier, and MOSFET pre-amplifier**—was carefully designed and integrated. **Analytical calculations for biasing, gain, impedance, and frequency response** were performed for every transistor stage, followed by **simulation-based verification** to ensure proper operation and performance compliance.

The **final combined amplifier** demonstrated **clean signal amplification** from a **millivolt-level microphone input** to a **speaker-level output**, achieving the **target gain of ≈ 100** , a **bandwidth of 100 Hz to 20 kHz**, and a **stable, low-distortion output**. All design specifications—including **output power, frequency response, and signal integrity**—were successfully met, confirming the effectiveness of the **multi-stage discrete design approach**. This project not only fulfilled the **technical objectives** but also reinforced a **practical understanding of transistor amplifier design, impedance matching, feedback stabilization, and systematic circuit integration**.